Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

Statement-1: The sum of $\cot^{-1}\left(\frac{7}{4}\right)+\cot^{-1}\left(\frac{19}{4}\right)+\ldots.+\cot^{-1}\frac{4r^2+3}{4}+\ldots.$ upto infinity is $\tan^{-1}2$.
Statement-2: $\sum_{r=1}^{n}\tan^{-1}\frac{x_r-x_{r-1}}{1+x_{r-1}x_r}=\tan^{-1}x_n-\tan^{-1}x_0$

(A) STATEMENT-1 isTrue, STATEMENT-2 is True; STATEMENT-2 is a correct explanation for STATEMENT-1

(B) STATEMENT-1 is True, STATEMENT-2 is True; STATEMENT-2 is NOT a correct explanation for STATEMENT-1

(C) STATEMENT-1 is True, STATEMENT-2 is False

(D) STATEMENT-1 is False, STATEMENT-2 is True

Q.2

$\theta=\tan^{-1}\left(2\tan^2\theta\right)-\tan^{-1}((1/3)\tan\theta)$ if $\tan\theta$ is equal to

(A) -2

(B) -1

(C) $2/3$

(D) 2

Q.3

If $\operatorname{cosec}^{-1}x=2\cot^{-1}7+\cos^{-1}(3/5)$ then the value of x is

(A) 44/117

(B) $125/117$

(C) $24/7$

(D) $5/3$

Q.4

If $\frac{1}{2}\sin^{-1}\left[\frac{3\sin 2\theta}{5+4\cos 2\theta}\right]=\tan^{-1}x$, then $x=$

(A) $\tan 3\theta$

(B) $3\tan\theta$

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(C) $(1/3)\tan\theta$

(D) $3\cot\theta$

Q.5

The sum of the infinite series $\cot^{-1}2+\cot^{-1}8+\cot^{-1}18+\cot^{-1}32+\ldots.$ is

(A) π

(B) $\frac{\pi}{2}$

(C) $\frac{\pi}{4}$

(D) None of these

Q.6

$\sum_{r=1}^{\infty}\tan^{-1}\left(\frac{1}{1+r+r^2}\right)=\ldots\ldots\ldots\ldots$

(A) $\frac{\pi}{2}$

(B) $\frac{\pi}{4}$

(C) $\frac{2\pi}{3}$

(D) None of these

Q.7

The sum of the infinite series
$\sin^{-1}\left(\frac{1}{\sqrt{2}}\right)+\sin^{-1}\left(\frac{\sqrt{2}-1}{\sqrt{6}}\right)+\sin^{-1}\left(\frac{\sqrt{3}-\sqrt{2}}{\sqrt{12}}\right)+\ldots$
$+\ldots+\sin^{-1}\left(\frac{\sqrt{n}-\sqrt{(n-1)}}{\sqrt{\{n(n+1)\}}}\right)+\ldots$ is

(A) $\frac{\pi}{8}$

(B) $\frac{\pi}{4}$

(C) $\frac{\pi}{2}$

(D) π

Q.8

The limit $\lim_{x\to\infty}x\left[\tan^{-1}\left(\frac{x+1}{x+2}\right)-\tan^{-1}\left(\frac{x}{x+2}\right)\right.$.
is equal to

(A) 2

(B) $\frac{1}{2}$

(C) $-\frac{1}{3}$

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(D) None of these

Q.9

The value of $tan^{-1}\frac{9}{19} + tan^{-1}\frac{9}{49} + tan^{-1}\frac{9}{97} + tan^{-1}\frac{9}{163} +$ ... ∞ equals

(A) $\tan^{-1}(3)$

(B) $\tan^{-1}\left(\frac{1}{3}\right)$

(C) $\tan^{-1}\left(\frac{2}{3}\right)$

(D) $\tan^{-1}\left(\frac{3}{2}\right)$

Q.10

Let $a = 2\tan^{-1}\sqrt{2} - 1$, $b = 3\sin^{-1}\frac{1}{\sqrt{2}} + \sin^{-1}-\frac{1}{2}$ and $c = \cos^{-1}\frac{1}{3}$. Then

(A) $a < b < c$

(B) $a < c < b$

(C) $c < b < a$

(D) $a > c > b$

Q.11

If $a\tan^{-1}e^{x} - b\tan^{-1}e^{-x} = c$, then value of $\cot^{-1}e^{x}$ is

(A) $\frac{a\pi+2c}{2(a+b)}$

(B) $\frac{a\pi-2c}{2(a+b)}$

(C) $\frac{a\pi+2c}{2(a-b)}$

(D) $\frac{a\pi-2c}{2(a-b)}$

Q.12

If $\sin^{-1}\left(x - \frac{x^2}{2} + \frac{x^3}{4} - \ldots\right)$
$+ \cos^{-1}\left(x^2 - \frac{x^4}{2} + \frac{x^6}{4} - \ldots\right) = \frac{\pi}{2}$ for $0 < x |< \sqrt{2}$
then x equals

(A) $1/2$

(B) 1

(C) $-1/2$

(D) -1

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Q.13

The value of $\cot^{-1} 7 + \cot^{-1} 8 + \cot^{-1} 18$ is

(A) π

(B) $\frac{\pi}{2}$

(C) $\cot^{-1} 5$

(D) $\cot^{-1} 3$

Q.14

The sum to the n term of the series
$\operatorname{cosec}^{-1} \sqrt{10} + \operatorname{cosec}^{-1} \sqrt{50} + \operatorname{cosec}^{-1} \sqrt{170} + \ldots$
$+ \operatorname{cosec}^{-1} \sqrt{\left(n^2+1\right)\left(n^2+2n+2\right)}$
is

(A) $\tan^{-1}(\mathrm{n}+1) - \pi/4$

(B) $\pi/4$

(C) $\tan^{-1}(\mathrm{n}+1)$

(D) 1

Q.15

The value of
$\sin^{-1}\left\{\cot\left(\sin^{-1}\sqrt{\left(\frac{2-\sqrt{3}}{4}\right)} + \cos^{-1}\frac{\sqrt{12}}{4} + \sec^{-1}\sqrt{2}\right)\right\}$ is

(A) 0

(B) $\frac{\pi}{1}$

(C) $\frac{\pi}{6}$

(D) $\frac{\pi}{2}$

Answers & Solutions

Q.1 Answer:

STATEMENT-1 isTrue, STATEMENT-2 is True; STATEMENT-2 is a correct explanation for STATEMENT-1

Solution:

L.H.S in Statement-2 is
$\sum_{r=1}^{n}\left(\tan^{-1} x_r - \tan^{-1} x_{r-1}\right) = \tan^{-1} x_n - \tan^{-1} x_0$
$\Rightarrow$ **Statement- 2 is true. In Statement-1, the given expression is equal to**

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$$\sum_{r=1}^{\infty}\tan^{-1}\frac{4}{4r^2+3}=\sum_{r=1}^{\infty}\tan^{-1}\frac{1}{1+(r^2-1/4)}$$
$$=\sum_{r=1}^{\infty}\tan^{-1}\frac{(r+1/2)-(r-1/2)}{1+(r+1/2)(r-1/2)}$$
$$=\lim_{n\to\infty}\left[\tan^{-1}(n+1/2)-\tan^{-1}(1/2)\right]$$

(using statement-2)

$=\pi/2-\tan^{-1}(1/2)=\cot^{-1}(1/2)=\tan^{-1}2$

So Statement- 1 is also true.

Q.2 Answer:

-2

Solution:

$\theta=\tan^{-1}\left(2\tan^2\theta\right)-\tan^{-1}((1/3)\tan\theta)$

$\Rightarrow\quad\tan\theta=\frac{2\tan^2\theta-(1/3)\tan\theta}{1+(2/3)\tan^3\theta}$

$\Rightarrow\quad\tan\theta\left[\frac{2\tan\theta-(1/3)}{1+(2/3)\tan^3\theta}-1\right]=0$

which is true if $\tan\theta=0$

or $\frac{2\tan\theta-(1/3)}{1+(2/3)\tan^3\theta}=1\Rightarrow\tan^3\theta-3\tan\theta+2=0$

$\Rightarrow\quad(\tan\theta-1)^2(\tan\theta+2)=0$ **which holds if** $\tan\theta=1$

or $\tan\theta=-2$.

Q.3 Answer:

$125/117$

Solution:

$2\cot^{-1}7+\cos^{-1}(3/5)=\cot^{-1}\frac{7^2-1}{2\times7}+\cot^{-1}\frac{3}{4}$

$\left[\because\text{ If }\theta=\cos^{-1}(3/5),\cos\theta=3/5,\cot\theta=3/4\right]$

$=\cot^{-1}(24/7)+\cot^{-1}(3/4)$

$=\cot^{-1}\left[\frac{\frac{24}{7}\times\frac{3}{4}-1}{\frac{24}{7}+\frac{3}{4}}\right].$

$=\cot^{-1}\frac{44}{117}=\operatorname{cosec}^{-1}\frac{125}{117}$

Q.4 Answer:

$(1/3)\tan\theta$

Solution:

$\frac{3\sin2\theta}{5+4\cos2\theta}=\frac{6\tan\theta}{9+\tan^2\theta}=\frac{2\tan\phi}{1+\tan^2\phi}=\sin2\phi$

where $\tan\theta=3\tan\phi$

$\Rightarrow\left(\frac{1}{2}\right)\sin^{-1}\left[\frac{3\sin2\theta}{5+4\cos2\theta}\right]=\phi=\tan^{-1}\left[\left(\frac{1}{3}\right)\tan\theta\right]$

so that $x=(1/3)\tan\theta$.

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Q.5 Answer:

$\frac{\pi}{4}$

Solution:

Let $S_\infty = \cot^{-1} 2 + \cot^{-1} 8 + \cot^{-1} 18 + \cot^{-1} 32 + \ldots$

$\therefore T_n = \cot^{-1} 2n^2$

$= \tan^{-1} \frac{1}{2n^2}$

$= \tan^{-1} \left(\frac{2}{4n^2}\right) = \tan^{-1} \left(\frac{(2n+1)-(2n-1)}{1+(2n+1)(2n-1)}\right)$

$= \tan^{-1}(2n+1) - \tan^{-1}(2n-1)$

$\therefore S_n = \sum_{n=1}^{\infty} \left\{\tan^{-1}(2n+1) - \tan^{-1}(2n-1)\right\}$

$= \tan^{-1} \infty - \tan^{-1} 1$

$= \frac{\pi}{2} - \frac{\pi}{4} = \frac{\pi}{4}$

Q.6 Answer:

$\frac{\pi}{4}$

Solution:

$\frac{1}{1+r+r^2} = \frac{1}{1+r(r+1)} = \frac{\frac{1}{r(r+1)}}{1+\frac{1}{r(r+1)}} = \frac{\frac{1}{r}-\frac{1}{r+1}}{1+\frac{1}{r}\cdot\left(\frac{1}{r+1}\right)}$

$\therefore \tan^{-1} \left(\frac{1}{1+r+r^2}\right) = \tan^{-1} \frac{1}{r} - \tan^{-1} \frac{1}{r+1}$

$\therefore \sum_{r=1}^{\infty} \tan^{-1} \left(\frac{1}{1+r+r^2}\right) = \tan^{-1} 1 = \frac{\pi}{4}$

Q.7 Answer:

$\frac{\pi}{2}$

Solution:

$\because \mathrm{T}_r = \sin^{-1} \left(\frac{\sqrt{r}-\sqrt{(r-1)}}{\sqrt{r(r+1)}}\right)$

$= \tan^{-1} \left(\frac{\sqrt{r}-\sqrt{(r-1)}}{1+\sqrt{r}\sqrt{(r-1)}}\right)$

$$S_n = \sum_{r=1}^{n} \tan^{-1} \left(\frac{\sqrt{r} - \sqrt{(r-1)}}{1 + \sqrt{r}\sqrt{(r-1)}}\right)$$

$= \tan^{-1} \sqrt{n} - \tan^{-1} \sqrt{0}$

$= \tan^{-1} \sqrt{n} - 0$

$\therefore \quad S_\infty = \tan^{-1} \infty = \frac{\pi}{2}$

Q.8 Answer:

$\frac{1}{2}$

Solution:

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$$\text{Lim}_{x\to\infty}\, x\left[\tan^{-1}\left(\frac{x+1}{x+2}\right)-\tan^{-1}\left(\frac{x}{x+2}\right)\right]$$
$$=\text{Lim}_{x\to\infty}\, x\tan^{-1}\left(\frac{\frac{x+1}{x+2}-\frac{x}{x+2}}{1+\frac{x+1}{x+2}\cdot\frac{x}{x+2}}\right)$$
$$=\text{Lim}_{x\to\infty}\, x\tan^{-1}\left(\frac{x+2}{2x^2+5x+4}\right)$$
$$=\text{Lim}_{x\to\infty}\, x\left(\frac{\tan^{-1}\left(\frac{x+2}{2x^2+5x+4}\right)}{\frac{x+2}{2x^2+5x+4}}\right)\times\frac{x(x+2)}{2x^2+5x+4}$$
$$=1\times\frac{1}{2}=\frac{1}{2}$$

Q.9 Answer:

$\tan^{-1}\frac{3}{2}$

Solution:

$$S=\lim_{m\to\infty}\sum_{n=1}^{m}tan^{-1}\frac{9}{9n^2+3n+7}=\lim_{m\to\infty}\sum_{n=1}^{m}tan^{-1}\frac{1}{1+n^2+\frac{n}{3}-\frac{2}{9}}$$
$$=\lim_{m\to\infty}\sum_{n=1}^{m}tan^{-1}\frac{n+\frac{2}{3}-n-\frac{1}{3}}{1+n+\frac{2}{3}n-\frac{1}{3}}=\lim_{m\to\infty}\sum_{n=1}^{m}tan^{-1}n+\frac{2}{3}-tan^{-1}n-\frac{1}{3}$$

Q.10 Answer:

$a < c < b$

Solution:

We have,

$a = 2\times\tan^{-1}\sqrt{2}-1=2\tan^{-1}\tan\frac{\pi}{8}=2.\frac{\pi}{8}=\frac{\pi}{4}=\cos^{-1}\frac{1}{\sqrt{2}}$

$b=3\sin^{-1}\frac{1}{\sqrt{2}}+\sin^{-1}-\frac{1}{2}=3\times\frac{\pi}{4}-\frac{\pi}{6}=\frac{7\pi}{12}$

$c=\cos^{-1}\frac{1}{3}$

Clearly $a < b$

Since $\frac{1}{3}<\frac{1}{\sqrt{2}}$

$\Rightarrow \cos^{-1}\frac{1}{3}>\cos^{-1}\frac{1}{\sqrt{2}}$ **(as** $\cos^{-1}x$ **decreasing in** $-1,1$**)**

$\Rightarrow c > a$

We observe that $0 < c < \frac{\pi}{2}$ **and** $\frac{\pi}{2} < b < \pi$

$\Rightarrow c < b$

Finally $a < c < b$

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Q.11 Answer:

$\frac{a\pi - 2c}{2(a+b)}$

Solution:

$a\frac{\pi}{2} - \cot^{-1}e^x - b\cot^{-1}e^x = c$

$\therefore \cot^{-1}e^x = \frac{a\pi - 2c}{2(a+b)}$

Q.12 Answer:

1

Solution:

$\sin^{-1}\left(x - \frac{x^2}{2} + \frac{x^3}{4} - \ldots\right) + \cos^{-1}\left(x^2 - \frac{x^4}{2} + \frac{x^6}{4} - \ldots\right) = \frac{\pi}{2}$

This is true only when $x - \frac{x^2}{2} + \frac{x^3}{4} - \ldots = x^2 - \frac{x^4}{2} + \frac{x^6}{4} \ldots\ldots$

$\Rightarrow \frac{x}{1+\frac{x}{2}} = \frac{x^2}{1+\frac{x^2}{2}}$

(Common ratios are $-\frac{x}{2}\&-\frac{x^2}{2}\&\mid$ **common ratios** $\mid < 1$**, in the given interval)**

$\frac{2x}{2+x} = \frac{2x^2}{2+x^2} \Rightarrow x = 0$ **or** $x = 1 \Rightarrow x = 1,$

$\{x$ **cannot be zero as** $0 < |x| < \sqrt{2}\}.$

Q.13 Answer:

$\cot^{-1}3$

Solution:

We have $\cot^{-1}7 + \cot^{-1}8 + \cot^{-1}18$

$\tan^{-1}\frac{1}{7} + \tan^{-1}\frac{1}{8} + \tan^{-1}\frac{1}{18}$

$= \tan^{-1}\left(\frac{\frac{1}{7}+\frac{1}{8}}{1-\frac{1}{7}\times\frac{1}{8}}\right) + \tan^{-1}\frac{1}{18}$

$= \tan^{-1}\frac{15}{55} + \tan^{-1}\frac{1}{18}\left(\because \frac{1}{7}\cdot\frac{1}{8} < 1\right)$

$\tan^{-1}\frac{3}{11} + \tan^{-1}\frac{1}{18} = \tan^{-1}\left(\frac{\frac{3}{11}+\frac{1}{18}}{1-\frac{3}{11}\times\frac{1}{18}}\right)$

$\left(\because \frac{3}{11}\cdot\frac{1}{18} < 1\right)$

$= \tan^{-1}\frac{65}{195} = \tan^{-1}\frac{1}{3} = \cot^{-1}3$

Q.14 Answer:

$\tan^{-1}(n+1) - \pi/4$

Solution:

Let $\theta = \operatorname{cosec}^{-1}\sqrt{(n^2+1)(n^2+2n+2)}$

$\Rightarrow \operatorname{cosec}^2\theta = (n^2+1)(n^2+2n+2)$

$= (n^2+1)^2 + 2n(n^2+1) + n^2 + 1$

$= (n^2+n+1)^2 + 1 \Rightarrow \cot^2\theta = (n^2+n+1)^2$

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$\Rightarrow \tan\theta = \frac{1}{n^2+n+1} = \frac{(n+1)-n}{1+(n+1)n}$

$\Rightarrow \theta = \tan^{-1}\left[\frac{(n+1)-n}{1+(n+1)n}\right] = \tan^{-1}(n+1) - \tan^{-1} n$

Thus, sum n terms of the given series $= \left(\tan^{-1} 2 - \tan^{-1} 1\right) + \left(\tan^{-1} 3 - \tan^{-1} 2\right)$
$+ \left(\tan^{-1} 4 - \tan^{-1} 3\right) + \ldots + \left(\tan^{-1}(n+1) - \tan^{-1} n\right)$

$\Rightarrow \tan^{-1}(n+1) - \pi/4$

Q.15 Answer:

0

Solution:

We have

$\sin^{-1}\left\{\cot\left(\sin^{-1}\sqrt{\left(\frac{2-\sqrt{3}}{4}\right)} + \cos^{-1}\frac{\sqrt{12}}{4} + \sec^{-1}\sqrt{2}\right)\right\}$

$= \sin^{-1}\left\{\cot\left(\sin^{-1}\sqrt{\left(\frac{\sqrt{3}-1}{2\sqrt{2}}\right)^2} + \cos^{-1}\frac{\sqrt{3}}{2} + \cos^{-1}\frac{1}{\sqrt{2}}\right)\right\}$

$= \sin^{-1}\left\{\cot\left(15^\circ + 30^\circ + 45^\circ\right)\right\} = \sin^{-1}\left(\cot 90^\circ\right)$

$= \sin^{-1} 0 = 0$

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